

PRAVIN KUMAR R

MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

Machine Learning Engineer with a Civil Engineering foundation (B.E., First Class) and expertise deploying end-to-end ML systems from research prototypes to production REST APIs. Built a 4-agent RAG orchestration system with deterministic Critic LLM evaluation gates, and a multi-architecture PyTorch forecasting pipeline achieving 2.06% MAPE under Walk-Forward Validation. Proficient in Docker, FastAPI, and GitHub Actions CI/CD, applying a production-first engineering mindset to GenAI, deep learning, and scalable applied ML systems.

SKILLS

MLOps & Deployment: Docker, FastAPI REST APIs, GitHub Actions CI/CD, Hugging Face Spaces, Python (production OOP), Git, SQLAlchemy

Deep Learning Systems: PyTorch (training loops, model serialization), PatchTST, LSTMs, Swin Transformer, CNNs, EfficientNet, ResNet, Hugging Face Transformers

GenAI & LLM Engineering: Groq LLaMA 3.3 70B, Multi-Agent RAG, DistilRoBERTa, TF-IDF Retrieval, LLM Evaluation Gates, Sentence-BERT, VADER Sentiment

ML Pipeline & Data: Scikit-Learn, Pandas, NumPy, SQL, Walk-Forward Validation, Cross-Validation, SHAP Explainability, Streamlit, Plotly, Feature Engineering

PROJECTS

ERIA: Multi-Agent RAG Orchestration System

Python, Groq LLaMA 3.3 70B, DistilRoBERTa, FastAPI, Docker, Hugging Face

- Architected a production 4-agent AI system (Analyst, Critic, Refiner, Strategist) powered by Groq LLaMA 3.3 70B, implementing a **deterministic Critic LLM evaluation gate** scoring each draft on grounding, consistency, and completeness across a 10-point quality rubric.
- Engineered a TF-IDF and cosine similarity Retrieval-Augmented Generation (RAG) pipeline parsing **50+ page policy documents** with contextual chunking, integrating a local DistilRoBERTa transformer for **7-class stakeholder emotion classification**.
- Deployed on Hugging Face Spaces with Docker containerization, FastAPI REST endpoints, and an automated PDF report generation pipeline at **\$0.0087 per run** under constrained compute budgets.

CryptoCast: Production ML Benchmarking & Deployment Pipeline

Python, PyTorch, PatchTST, LSTM, Streamlit, SHAP, GitHub Actions

- Designed a centralized PyTorch model training framework benchmarking **5 deep learning architectures** (1D-CNN, RNN, LSTM, Transformer, PatchTST) under chronological Walk-Forward Validation across ~5,000 daily records, enforcing production-grade model selection discipline.
- Selected the optimal architecture achieving **2.06% MAPE at 1-day** and 5.93% MAPE at 7-day horizons; engineered protocol supply features (Halving cycles, Block Reward) driving a **\$92 MAE reduction** on 7-day ablation benchmarks.
- Delivered an end-to-end production pipeline with **GitHub Actions CI**, automated model checkpointing, and an 11-page Streamlit analytics studio with SHAP explainability — containerized for repeatable, version-controlled inference deployment.

Other Verified Projects: HuBMAP Swin-TransUNet (0.6782 Dice, +26.7%), Alveris-Geo-AI (95.98% Accuracy, Basel III Climate VaR), Soil Grain Size Estimation (55.22 OOF Weighted EMD, Elastic Net) – on GitHub.

PROFESSIONAL EXPERIENCE

Data Science Intern — Internship Studio (Analytics & EdTech)

Jan 2026 – Feb 2026 | Remote

- Analyzed **14,102 Fitbit telemetry records**, executing feature engineering, categorical encoding, and StandardScaler normalization under an 80/20 stratified train-test split, achieving **99.8% feature completeness**.
- Engineered a Random Forest regression model achieving **$R^2 = 0.9979$ and MAE of 3.78 kcal** (89.2% error reduction over baseline), containerizing the trained model with Docker and serving predictions via FastAPI REST endpoints.
- Segmented users into 4 exertion zones using PCA and K-Means clustering (**Silhouette 0.4546**), deploying the full ML pipeline with GitHub Actions CI — cutting manual validation turnaround time by **75%**.

EDUCATION

Master Data Science Program — HCL GUVI, IIT Madras Research Park Incubated

Completed Jun 2026

B.E. Civil Engineering (First Class Honors) — Rajalakshmi Engineering College, Anna University

Aug 2017 – May 2021

CERTIFICATIONS

Certifications: Data Science & Analytics (HCL GUVI), Build & Deploy AI Apps with Google AI Studio, ML for Earth Observation (EO College), Machine Learning & Pandas (HCL GUVI)